

Journal Club

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Dreaming outside the Box: Evidence for Memory Abstraction in REM Sleep

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Review of Pereira et al.

A likely function of sleep is to reorganize neural information, allowing us to make new associations and find patterns of meaning from past experiences. Rapid eye movement sleep (REM) has long been thought to play a role in this process, largely because of the immersive, bizarre, and at times metaphorical dreams that occur in this state. Experimental studies offer preliminary evidence for this connection. For instance, time in REM enhanced creativity and recombining of schemas (Landmann et al., 2015). Associative thinking also increased after waking from REM relative to non-rapid eye movement sleep (NREM) (Landmann et al., 2015). These studies and others suggest a link between REM and processes related to abstraction.

Whether REM plays any role in memory consolidation, however, is a topic of controversy. Pathologies and medications that reduce REM do not lead to obvious cognitive deficits, and studies that specifically deprive participants from REM introduce confounds, such as stress. Yet,

qualitative aspects of memory consolidation, such as abstraction, may be missed by these assessments. Indeed, it is challenging to determine how best to operationalize abstraction, which involves creating and updating schemas often specific to each individual's experience. Stronger conclusions could be made from measures that capture the essence of abstraction, evaluating the degree to which patterns are accurately distilled from a corpus of information. Further, many studies report correlations between time in REM and outcomes but without manipulating memory processing during sleep, they leave open the possibility that other variables drive the connection.

A procedure known as targeted memory reactivation has played a pivotal role in understanding the causal involvement of sleep in learning. In this procedure, sounds or smells are presented during learning and then again during sleep to trigger processing of specific associated memories. When applied in NREM, targeted memory reactivation reliably boosts subsequent memory retrieval on a variety of behavioral tasks (Hu et al., 2020). Many experiments have applied targeted memory reactivation to REM, but they often result in null or moderate impacts on learning (Hu et al., 2020). This lack of consistency has led researchers to develop increasingly complex and ecological tasks, revealing intriguing effects of reactivating memories in REM. With a face recognition task, targeted memory reactivation during REM increased both accurate

recall and false alarms, suggesting that it may have promoted generalization (Sterpenich et al., 2014). However, this study could not rule out the possibility that targeted memory reactivation generally increased feelings of recognition rather than generalization of previously seen faces. Thus, the jury is still out on whether targeted memory reactivation specifically boosts generalization.

Another intriguing aspect of REM is the time course of learning effects. Indeed, it has been suggested that the complex reorganization of information inherent in consolidation may take days or weeks to unfold. In particular, REM has been hypothesized to play a role in slower plasticity events, such as myelination and memory corticalization, which are relevant for forming schemas (Pereira and Lewis, 2020). The time required for this complex NREM-REM interplay may explain the lack of consistency in studies linking REM and learning, as most testing occurs immediately after sleep.

Through the use of a novel abstraction task and delayed testing, Pereira et al. (2023) took an impressive step forward in testing the hypothesized role of REM in abstraction. They used targeted memory reactivation to reactivate parts of an abstraction task in both NREM and REM within the same night to assess its effect on task performance up to a week later. Abstraction was measured using the Synthetic Visual Reasoning Task (SVRT), a statistical learning

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task originally developed to test the abstract reasoning of artificial intelligence. The SVRT consists of simple line drawings following hidden rules that allow for categorization (e.g., two of four shapes must be identical). In the current study, there were 16 different sets of images with hidden rules, referred to as problems. For each problem, participants saw a sample image following the rule and learned via trial and error to identify whether novel images fit the rule. Each rule was associated with a distinctive sound that could then be used as a targeted memory reactivation cue during sleep.

Half the problems were reactivated during sleep: four in NREM and four in REM. Control sounds (not associated with any problems) were also presented, allowing the authors to confirm with ERPs and slow-oscillation analyses that targeted memory reactivation cues were distinctively processed during both stages. In the morning and again 1 week later, participants determined whether or not five new images for each problem followed the rule. Performance at the week delay was better for problems cued during REM than for uncued problems and NREM-cued problems. This result supports the twofold claim that REM uniquely facilitates rule abstraction and that it takes more than one night to do so.

The delayed emergence of a REM-cueing effect provides novel evidence for the idea that reactivating abstraction problems in REM taps into slower plasticity processes. In their sequential model of memory processing, *Pereira and Lewis (2020)* suggested that memories are tagged during wake, and then memory tags are “captured” at synapses and stabilized during NREM reactivation. In subsequent REM, captured tags are translated into proteins, enabling plasticity and synaptic remodeling to unfold over multiple REM periods in the subsequent days. This complex NREM-REM interplay might explain why some studies find that NREM cueing has a greater benefit for memory integration when followed by REM. However, in the *Pereira et al. (2023)* study, there was no effect of presenting cues in NREM the next morning or at a delay. One explanation proposed by *Pereira et al. (2023)* is that reactivating memories in REM might have blocked the NREM-REM interplay that would otherwise have occurred. However, this seems not to be the case since NREM cues did not improve abstraction even for participants who had ample time in subsequent uncued REM. Replicating this finding with a between-subjects design or over multiple nights would help determine whether

or not cues interfere with processes that would otherwise unfold naturally.

Alternatively, the delayed emergence of a REM-cueing effect could suggest that reactivation in REM did not just replace NREM reactivation in capturing memory tags, but tapped into processes unique to REM that are crucial for facilitating abstraction. Whereas NREM reactivations are thought to arise from the hippocampus and promote the transfer of memories to neocortical sites, the neurochemical milieu in REM, including high acetylcholine and low norepinephrine, may promote spreading activation within the neocortex with less influence from the hippocampus (*Diekelmann and Born, 2010*). Thus, perhaps targeted memory reactivation in REM is uniquely capable of influencing cortical reactivations, which may be integral for the type of information restructuring that promotes rule abstraction. REM reactivation could be further facilitated by the hyperassociative nature of REM cognition, enabling targeted memory reactivation to reactivate more distantly related information. Although the authors did not collect dream reports in this study, their delayed learning effect is consistent with the dream-lag effect, whereby waking life events incorporate in REM dreams with a 5–7 d delay (*Blagrove et al., 2011*). Targeted memory reactivation in REM and NREM also produces differential delayed effects on dream content over a week (*Picard-Deland and Nielsen, 2022*), which again supports the idea that memory processing unfolds with a distinct time course depending on the sleep stage in which it was reactivated.

Further investigating dream content during and after reactivation of an abstraction task may yield insight into the psychological correlates or mechanisms of targeted memory reactivation. For instance, information reactivated during REM may trigger metaphorically related dream content that predicts gains in abstraction. A recent meta-analysis found that performance improves more on a task that is dreamt about in NREM compared with REM (*Hudachek and Wamsley, 2023*). Finding ways to measure not only whether dreams incorporate a task, but also how they diverge from waking experiences may help better understand REM's function in memory processing and provide a window into the abstraction process. Dream engineering, using methods to modify dream content, such as sensory stimulation, could also be used to test causal relationships between dreaming and behavioral changes (*Carr et al., 2020*). For instance, the

Pereira et al. (2023) study design could be replicated in a population of lucid dreamers, where cues could be used to request dreams of specific abstraction problems.

Rule abstraction assessed via the SVRT is just one facet of how memories may transform during REM. Continuing to identify tasks sensitive to REM reactivation will enable further progress in determining the functions of REM. Further, the development of high-resolution at-home wearable EEG devices will make extended sleep studies more feasible, allowing us to better assess long-term REM effects, which may otherwise be missed by single-night sleep studies. As we continue improving our ability to measure elusive aspects of memory transformation, it will be easier to detect and understand the functions of REM.

References

- Blagrove M, Fouquet NC, Henley-Einion JA, Pace-Schott EF, Davies AC, Neuschaffer JL, Turnbull OH (2011) Assessing the dream-lag effect for REM and NREM stage 2 dreams. *PLoS One* 6:e26708.
- Carr M, Haar A, Amores J, Lopes P, Bernal G, Vega T, Rosello O, Jain A, Maes P (2020) Dream engineering: simulating worlds through sensory stimulation. *Conscious Cogn* 83:102955.
- Diekelmann S, Born J (2010) The memory function of sleep. *Nat Rev Neurosci* 11:114–126.
- Hu X, Cheng LY, Chiu MH, Paller KA (2020) Promoting memory consolidation during sleep: a meta-analysis of targeted memory reactivation. *Psychol Bull* 146:218–244.
- Hudachek L, Wamsley EJ (2023) A meta-analysis of the relation between dream content and memory consolidation. *Sleep*. Advance online publication. Retrieved Apr 14, 2023. <https://doi.org/10.1093/sleep/zsad111>.
- Landmann N, Kuhn M, Maier JG, Spiegelhalter K, Baglioni C, Frase L, Riemann D, Sterr A, Nissen C (2015) REM sleep and memory reorganization: potential relevance for psychiatry and psychotherapy. *Neurobiol Learn Mem* 122:28–40.
- Pereira SI, Lewis PA (2020) The differing roles of NREM and REM sleep in the slow enhancement of skills and schemas. *Curr Opin Physiol* 15:82–88.
- Pereira SI, Santamaria L, Andrews R, Schmidt E, Rossum MC, Lewis P (2023) Rule abstraction is facilitated by auditory cuing in REM sleep. *J Neurosci* 43:3838–3848.
- Picard-Deland C, Nielsen T (2022) Targeted memory reactivation has a sleep stage-specific delayed effect on dream content. *J Sleep Res* 31:e13391.
- Sterpenich V, Schmidt C, Albouy G, Matarazzo L, Vanhaudenhuyse A, Boveroux P, Degueldre C, Leclercq Y, Balet E, Collette F, Luxen A, Phillips C, Maquet P (2014) Memory reactivation during rapid eye movement sleep promotes its generalization and integration in cortical stores. *Sleep* 37:1061–1075.